Module Interface Specification for CEWS System

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1 Revision History

Date	Version	Notes
2022-03-10	0.0	Creation
2022-03-17	1.0	Version 1.0

2 Symbols, Abbreviations and Acronyms

See SRS Documentation at https://github.com/LesleyWheat/CAS741_CEWS/blob/main/docs/SRS/SRS.pdf .

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3 Introduction

The following document details the Module Interface Specifications for CEWS System. This program implements the Cut Edge Weight Statistic measurement, proposed by Zighed et al. (1). Complementary documents include the System Requirement Specifications and Module Guide. The full documentation and implementation can be found at https://github.com/LesleyWheat/CAS741_CEWS/blob/main/docs/SRS/SRS.pdf.

4 Notation

The structure of the MIS for modules comes from (2), with the addition that template modules have been adapted from (3).

The following table summarizes the primitive data types used by CEWS System.

Data Type	**Notation**	**Description**
natural number	$\mathbb{N}$	a number without a fractional component in $[1, \infty)$
real	$\mathbb{R}$	any number in $(-\infty, \infty)$
matrix	$\mathbb{M}_{nxm}$	a matrix of real numbers of size n and m

The specification of CEWS System uses some derived data types: sequences, strings, and tuples. Sequences are lists filled with elements of the same data type. Strings are sequences of characters. Tuples contain a list of values, potentially of different types. In addition, CEWS System uses functions, which are defined by the data types of their inputs and outputs. Local functions are described by giving their type signature followed by their specification.

5 Module Decomposition

The following table is taken directly from the Module Guide document for this project.

Level 1	**Level 2**
Hardware-Hiding Module	
Behaviour-Hiding Module	Control Module Input Module Output Module Calculation Module Graph Module
Software Decision Module	

Table 1: Module Hierarchy

6 MIS of Control Module

6.1 Module

control

6.2 Uses

Input Module 7, Calculation Module 9, Output Module 8

6.3 Syntax

6.3.1 Exported Constants

None.

6.3.2 Exported Access Programs

Name	In	Out	Exceptions
cutEdgeWeight	inData: $\mathbb{M}_{nxm}$, inLabels: $\mathbb{M}_{nx1}$	J_N: $\mathbb{R}$	-

6.4 Semantics

6.4.1 State Variables

None.

6.4.2 Environment Variables

None.

6.4.3 Assumptions

- The inputs are valid.
- The input data has been formatted correctly.

6.4.4 Access Routine Semantics

cutEdgeWeight():

- output: J_N computed by calculation module 9.

  Controls the information flow.

- Pass input to Input Module 7

- Pass result to Calculation Module 9, receive J_N
- Pass J_N to Output Module 8 for verification

6.4.5 Local Functions

None.

7 MIS of Input Module

7.1 Module

input

7.2 Uses

None.

7.3 Syntax

7.3.1 Exported Constants

None.

7.3.2 Exported Access Programs

Name	**In**	**Out**	**Exceptions**
processInput	inData: $\mathbb{M}_{nxm}$, inLabels: $\mathbb{M}_{nx1}$	data: $\mathbb{M}_{nxm}$, labels: $\mathbb{M}_{nx1}$	typeError formatError

7.4 Semantics

7.4.1 State Variables

None.

7.4.2 Environment Variables

None.

7.4.3 Assumptions

- The input data has been formatted correctly (for example, that inData[i] is the data point associated with inLabels[i]).

7.4.4 Access Routine Semantics

processInput():

- transition: Verify and transform input if applicable
  - Check if input is a correct data type
  - Check input meets constraints
  - Convert information to internal data programming object

- output:
  - data: the inputted data converted to the correct internal data storage type for use by the Calculation Module 9
  - labels: the inputted labels converted to the correct internal data storage type for use by the Calculation Module 9
- exceptions:
  - typeError: inData or inLabels are not accepted data objects (for example, a list of strings).
  - formatError:
    * inData and inLabels have different values of n
    * Different rows of inData have a different m value
    * Any index inData or inLabels is empty

7.4.5 Local Functions

None.

8 MIS of Output Module

8.1 Module

output

8.2 Uses

None.

8.3 Syntax

8.3.1 Exported Constants

None.

8.3.2 Exported Access Programs

Name	In	Out	Exceptions
verifyOutput	J_N: $\mathbb{R}$	-	resultOutOfBounds

8.4 Semantics

8.4.1 State Variables

None.

8.4.2 Environment Variables

None.

8.4.3 Assumptions

- J_N comes from the Calculation Module 9.

8.4.4 Access Routine Semantics

verifyOutput():

- exception:
  - resultOutOfBounds: If $J_N < 0$ or $J_N > 1$.

8.4.5 Local Functions

None.

9 MIS of Calculation Module

9.1 Module

calc

9.2 Uses

Graph Module 10

9.3 Syntax

9.3.1 Exported Constants

None.

9.3.2 Exported Access Programs

Name	**In**	**Out**	**Exceptions**
calcCEWS	data: $\mathbb{M}_{nxm}$ labels: $\mathbb{M}_{nx1}$	J_N: $\mathbb{R}$	-

9.4 Semantics

9.4.1 State Variables

None.

9.4.2 Environment Variables

None.

9.4.3 Assumptions

- Input variables have been processed through Input Module 7.

9.4.4 Access Routine Semantics

calcCEWS():

- output:
  - The Cut Edge Weight Statistic: J_N given by $\frac{J}{I+J}$ where $J = \sum_{r=1}^{k-1} \sum_{s=r+1}^{k} J_{r,s}$; $0 \leq J \leq n^2$ where each $J_{r,s} = \sum_{i=1}^{n} \sum_{j=1}^{n} V_{ij} \times I(c_i = r \ \& \ c_j = s)$ and $I = \sum_{r=1}^{k} I_r, 0 \leq I \leq n^2$ where each $I_r = \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} V_{ij} \times I(c_i = r \ \& \ c_j = r)$. V is obtained from the Graph Module 10.

9.4.5 Local Functions

None.

10 MIS of Graph Module

10.1 Module

graph

10.2 Uses

None.

10.3 Syntax

10.3.1 Exported Constants

None.

10.3.2 Exported Access Programs

Name	In	Out	Exceptions
relativeNeighbourGraph	data: $\mathbb{M}_{nxm}$	V: $\mathbb{M}_{nxn}$	-

10.4 Semantics

10.4.1 State Variables

None.

10.4.2 Environment Variables

None.

10.4.3 Assumptions

- data has been processed by Input Module 7.

10.4.4 Access Routine Semantics

relativeNeighbourGraph():

- transition:
  - Compute distance matrix
  - Construct graph
- output:

– Connection matrix $V = (v_{ij}), i = 1, 2, ..., n; j = 1, 2, ..., n$;
where $v_{ij} = I(E_{RNG}(P, i, j))$ as per: $E_{RNG}(P, i, j) = [d(p_i, p_j) \leq \max(d(p_i, c), d(p_j, c)) \forall c$ where $c \in P, c \neq p_i, c \neq p_j]$.

10.4.5 Local Functions

None.

References

[1] D. A. Zighed, S. Lallich, and F. Muhlenbach, "A statistical approach to class separability," *Applied Stochastic Models in Business and Industry*, vol. 21, no. 2, pp. 187–197, Mar. 2005. [Online]. Available: https://onlinelibrary.wiley.com/doi/10.1002/asmb.532

[2] D. M. Hoffman and P. A. Strooper, *Software Design, Automated Testing, and Maintenance: A Practical Approach*. New York, NY, USA: International Thomson Computer Press, 1995. [Online]. Available: http://citeseer.ist.psu.edu/428727.html

[3] C. Ghezzi, M. Jazayeri, and D. Mandrioli, *Fundamentals of Software Engineering*, 2nd ed. Upper Saddle River, NJ, USA: Prentice Hall, 2003.